# Artificial Intelligence

INTRODUCTION



# Artificial Intelligence

- Defined as anything that makes machines act more intelligent
- Intelligence: “The  capacity to learn and solve problems”
- Artificial Intelligence simulation of human intelligence
  - The ability to solve problems
  - The ability to act rationally
  - The ability to act like a humans

# Artificial Intelligence

"It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions."

# Artificial Intelligence

AI is the ability of a machine to perform tasks that are typically associated with human intelligence, such as learning, problem-solving, language learning, and decision-making.

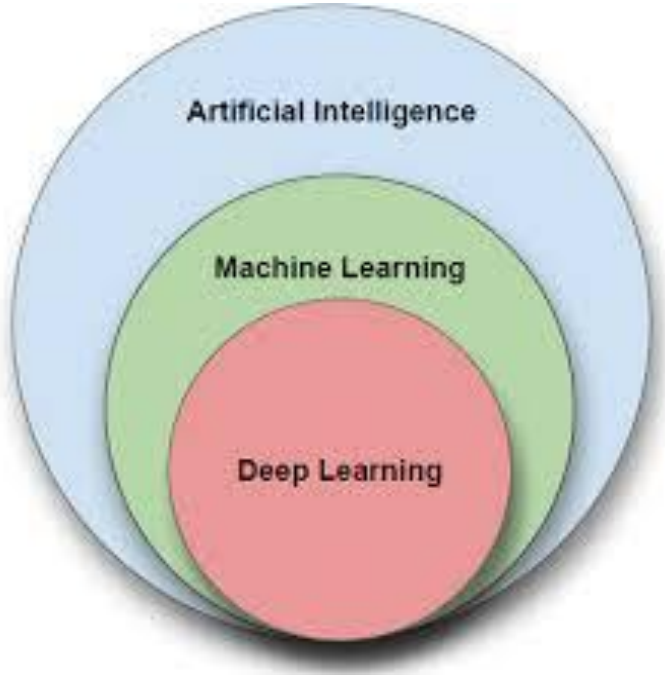
AI systems aim to mimic human cognitive abilities

John McCarthy  coined the term “artificial intelligence”

# Artificial Intelligence

AI is the science of building intelligent machines from large volumes of data and learning from experience to perform human-like tasks.

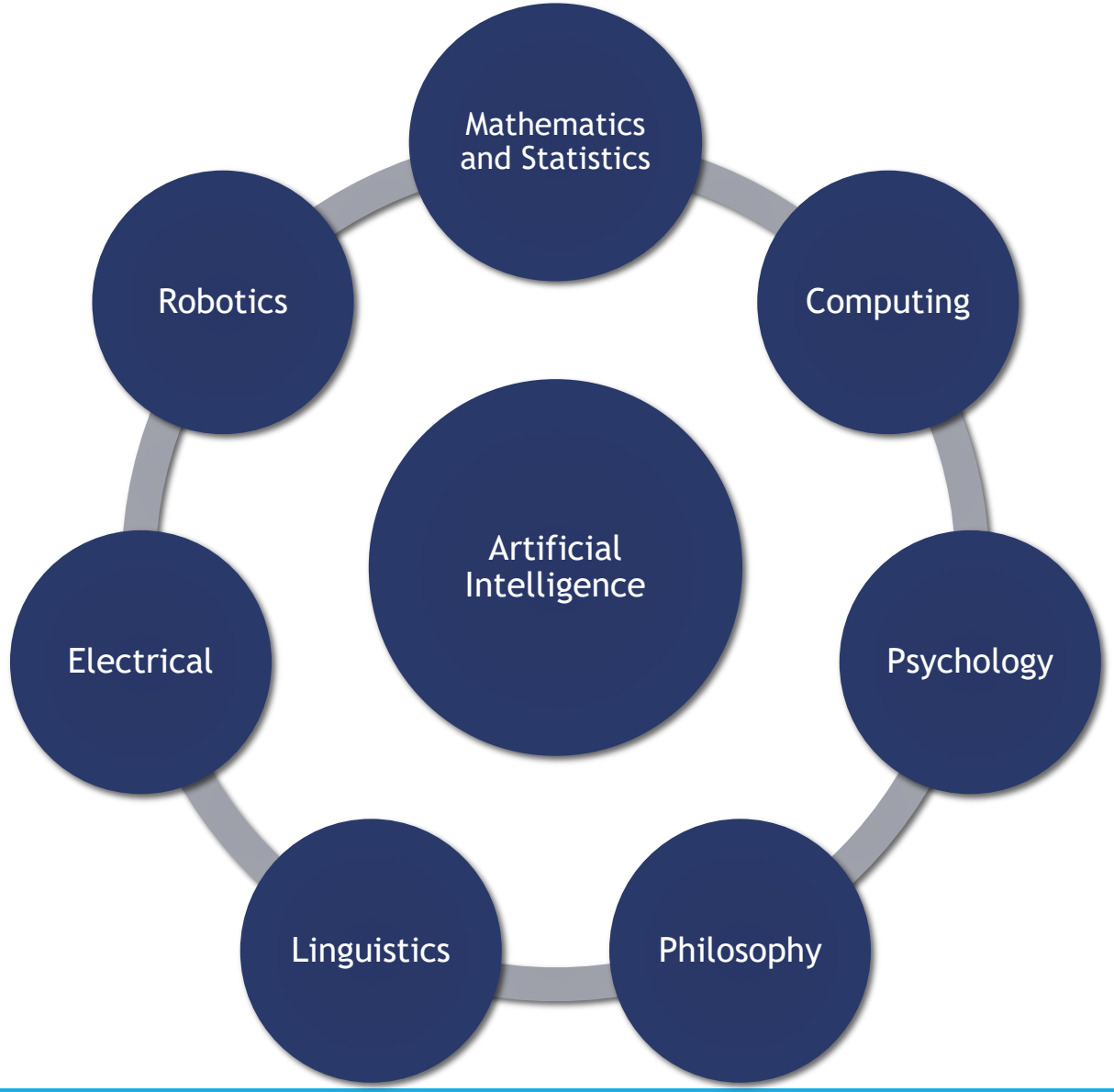


# Why Artificial Intelligence?

- you can create such software or devices : health issues, marketing, traffic issues, etc.
- you can create your personal virtual Assistant: Cortana, Google Assistant, Siri, etc.
- you can build such Robots which can work in an environment where survival of humans can be at risk.
- AI opens a path for other new technologies, new devices, and new Opportunities.

# AIML is a multi-disciplinary field



# Applications of AI

# Applications

- Smartphones
- Security and Surveillance
- Social Media Platforms
- Navigation
- E-commerce
- Banking and Finance Sector
- Autonomous Vehicle
- Smart Home

# Smart Compose **of GMAIL**

Using Neural Networks to Help Write Emails

Hey Snehal,
Haven’t seen you in a while and I hope you’re doing well

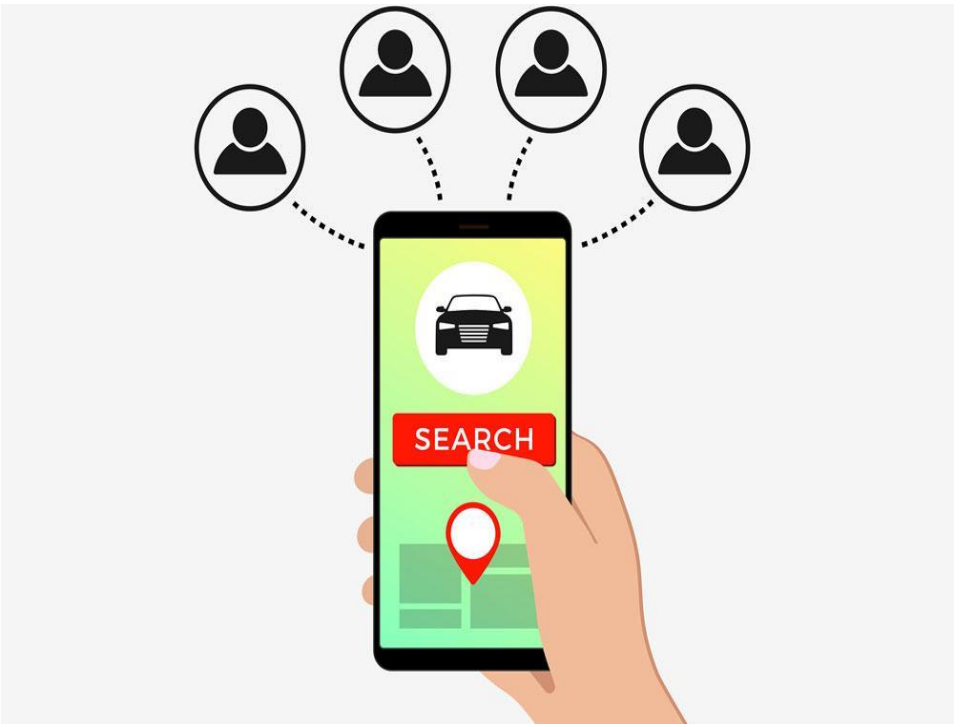
# Spam Filters and Smart Email Categorization



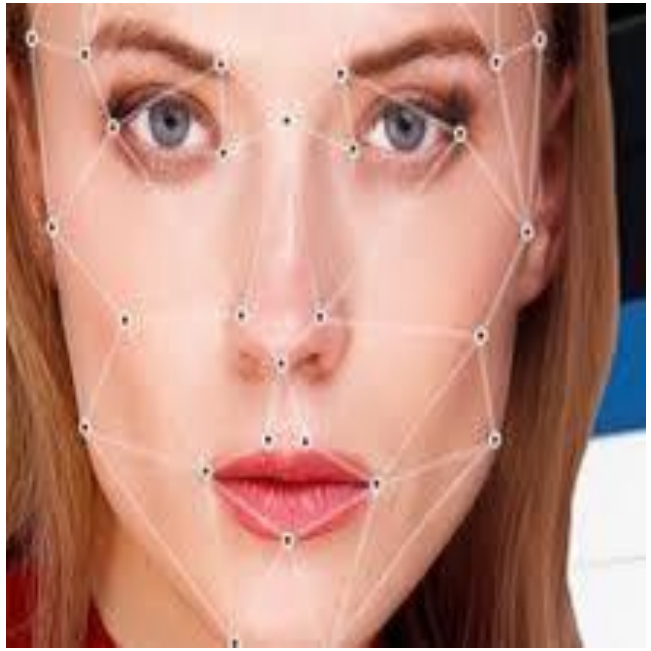



# Ridesharing Apps Like UBER and OLA



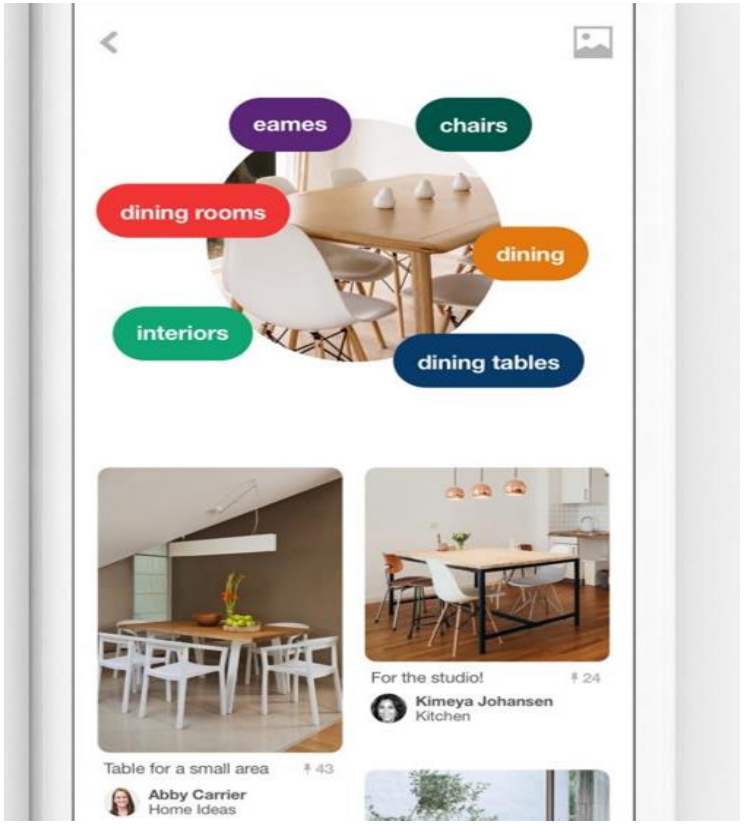
# Social Networking





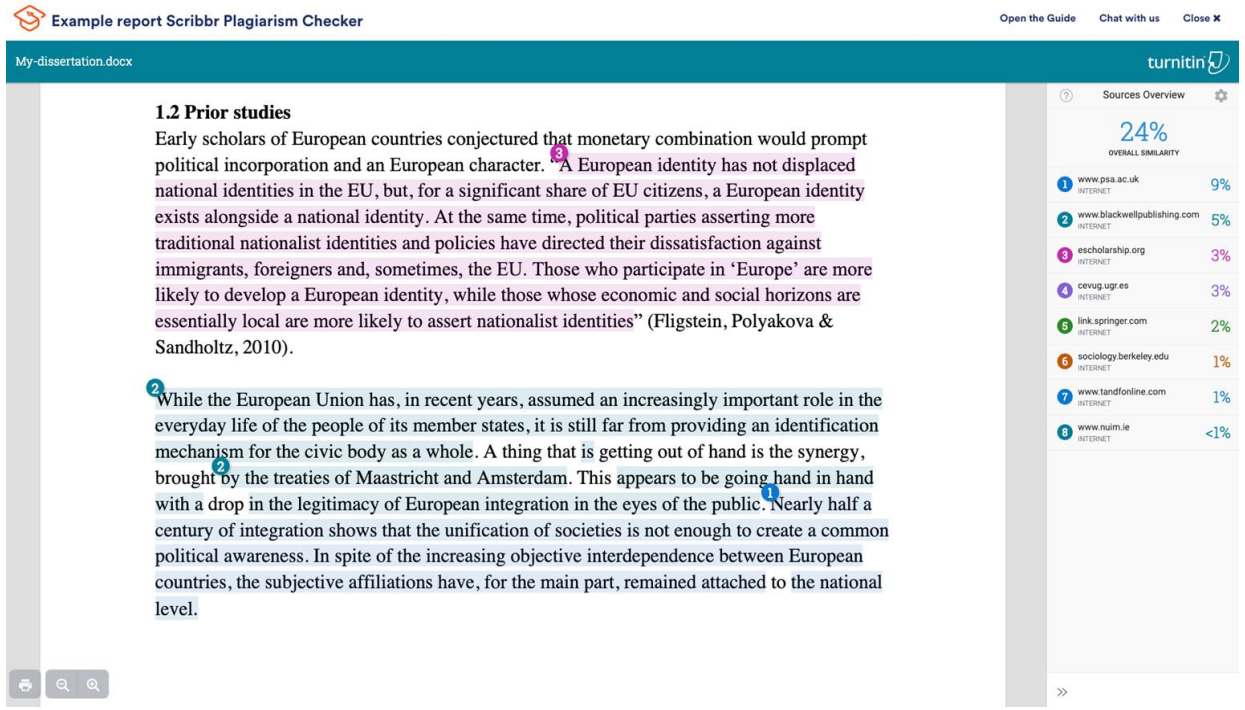


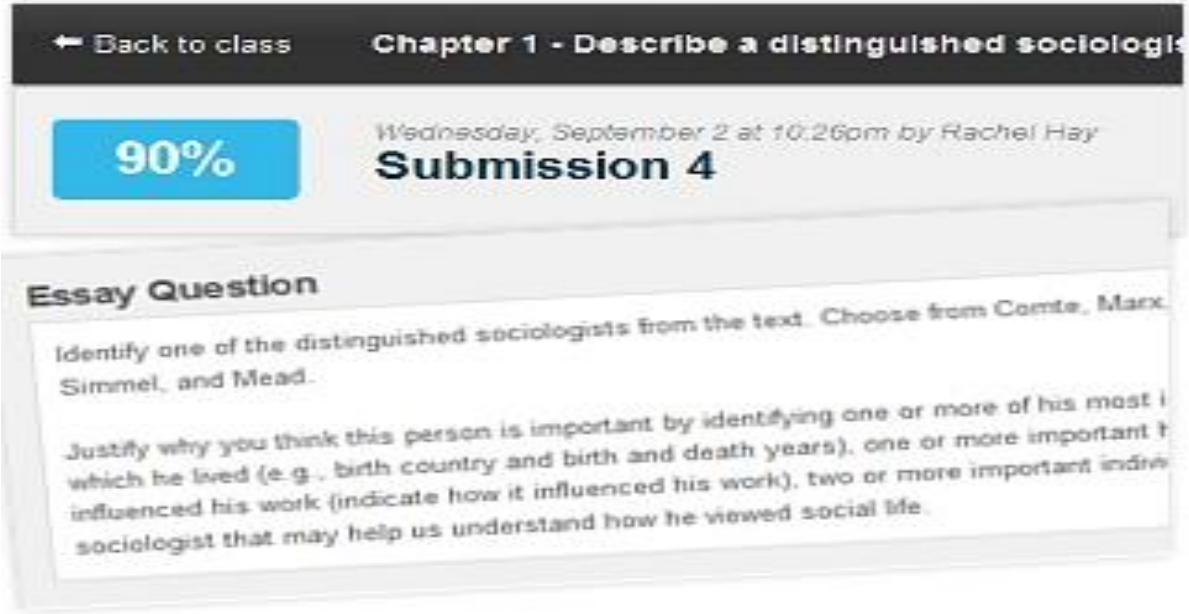


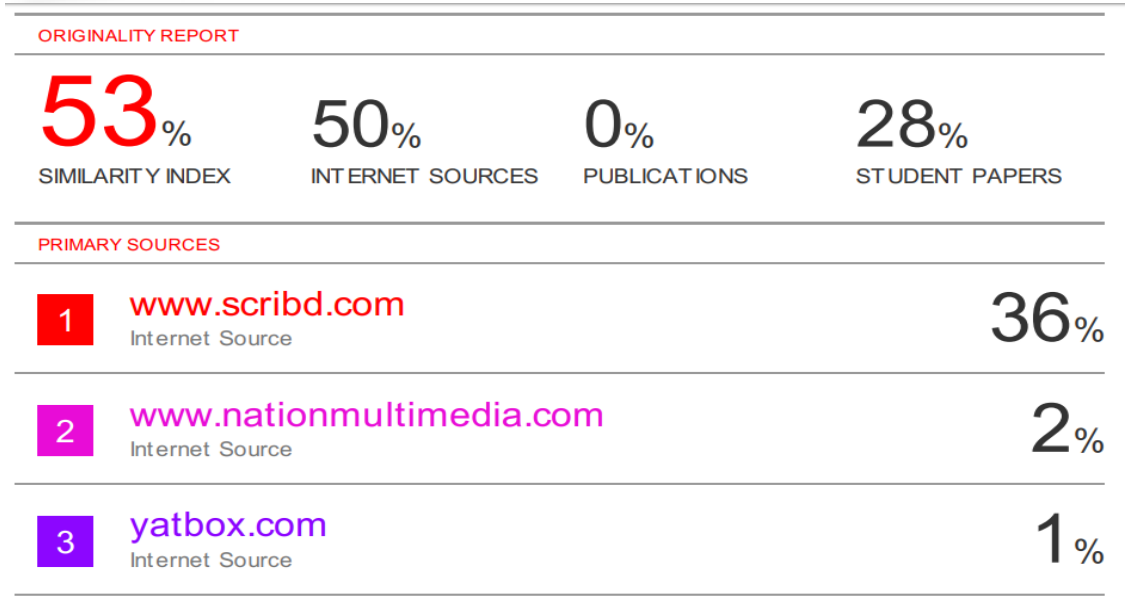


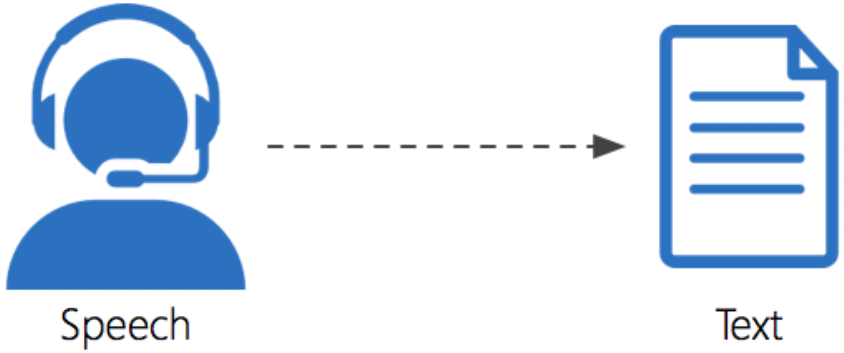
# Grading and Assessment

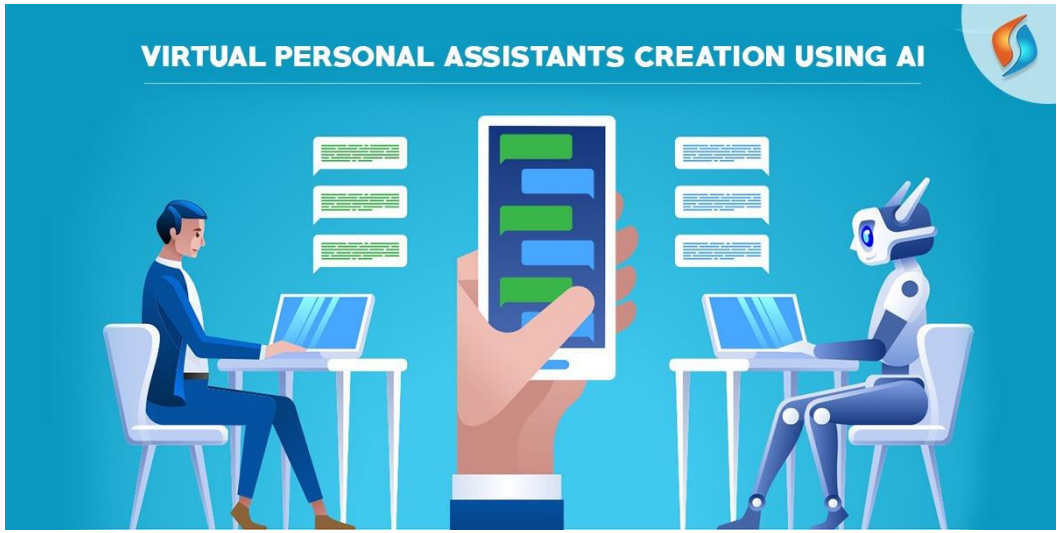
## Plagiarism Checkers Robo-readers

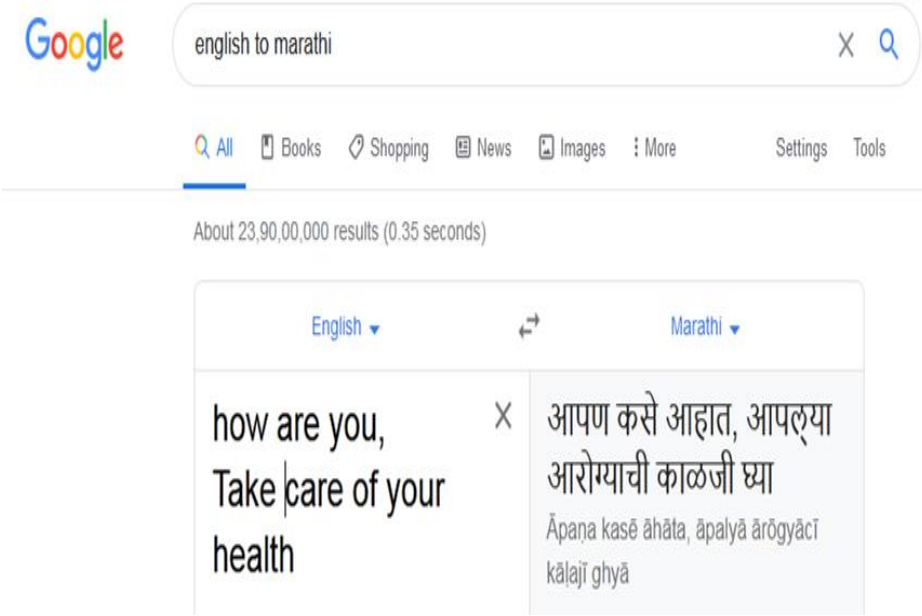


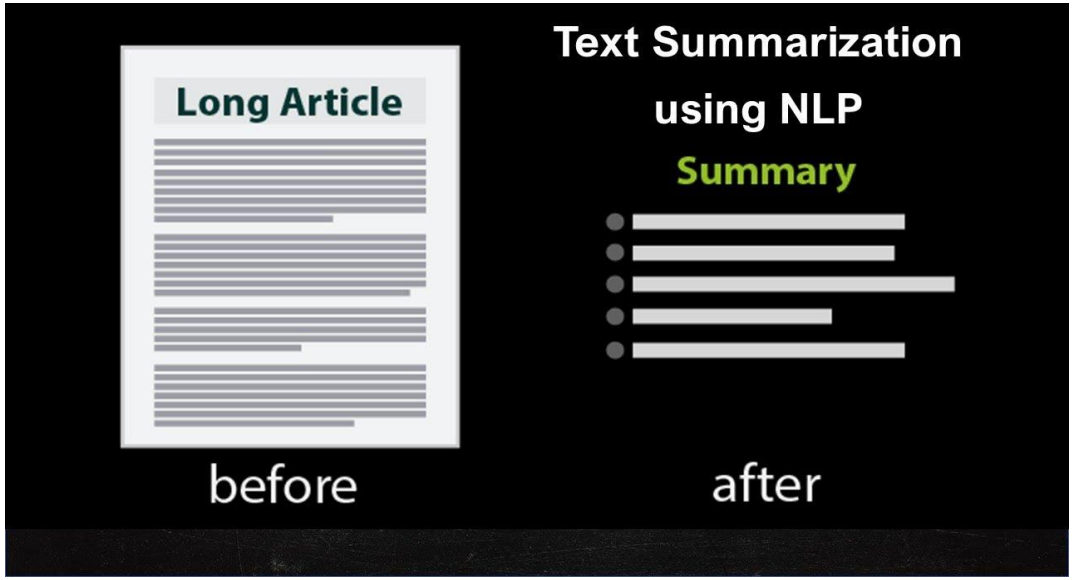




Speech
Text


VIRTUAL PERSONAL ASSISTANTS CREATION USING AI


Google
english to marathi
All
Books
Shopping
News
Images
More
Settings
Tools
About 23,90,00,000 results (0.35 seconds)
English
Marathi
how are you, Take care of your health
आपण कसे आहात, आपल्या आरोग्याची काळजी घ्या
Āpaṇa kasē āhāta, āpalyā ārōgyācī kāḷajī ghyā


Long Article
Text Summarization using NLP
Summary
before
after

# Self Driving Car

https://www.landmarkdividend.com/self-driving-car/



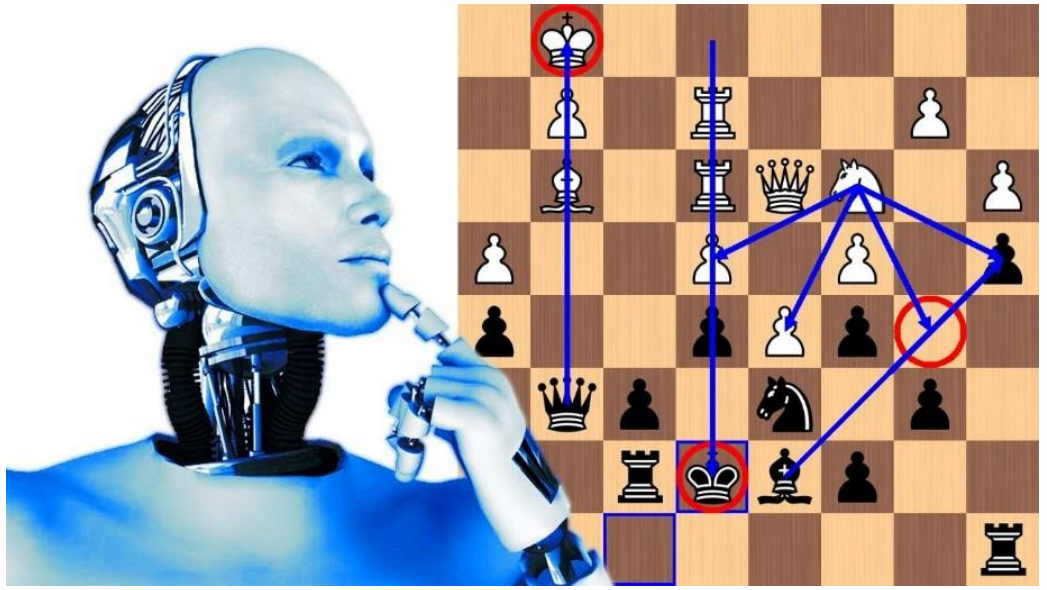



# Commercial Flights Use an AI Autopilot



# Deep Blue (chess computer)



# AlphaGo

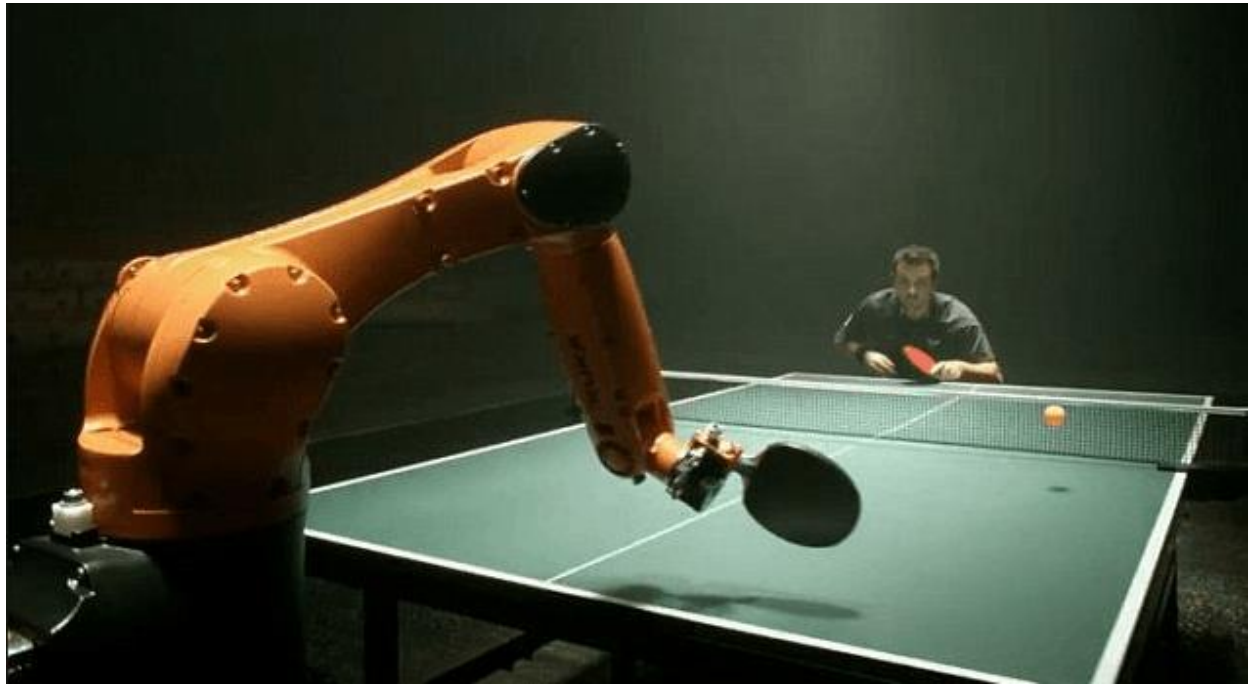


# Robo Soccer



# The Duel: Timo Boll vs. KUKA Robot

https://spectrum.ieee.org/robots-playing-ping-pong-whats-real-and-whats-not#toggle-gdpr



https://www.youtube.com/watch?v=tIIJME8-au8

# 1. To Demonstrate Real-World Use of AI

These examples visualize how AI works:

AlphaGo → shows how AI can learn and make intelligent decisions.

Robo Soccer → shows teamwork, planning, and perception in robotics.

KUKA Robot Duel → shows AI-powered motion control and precision.

# 2. To Illustrate Key AI Concepts Practically

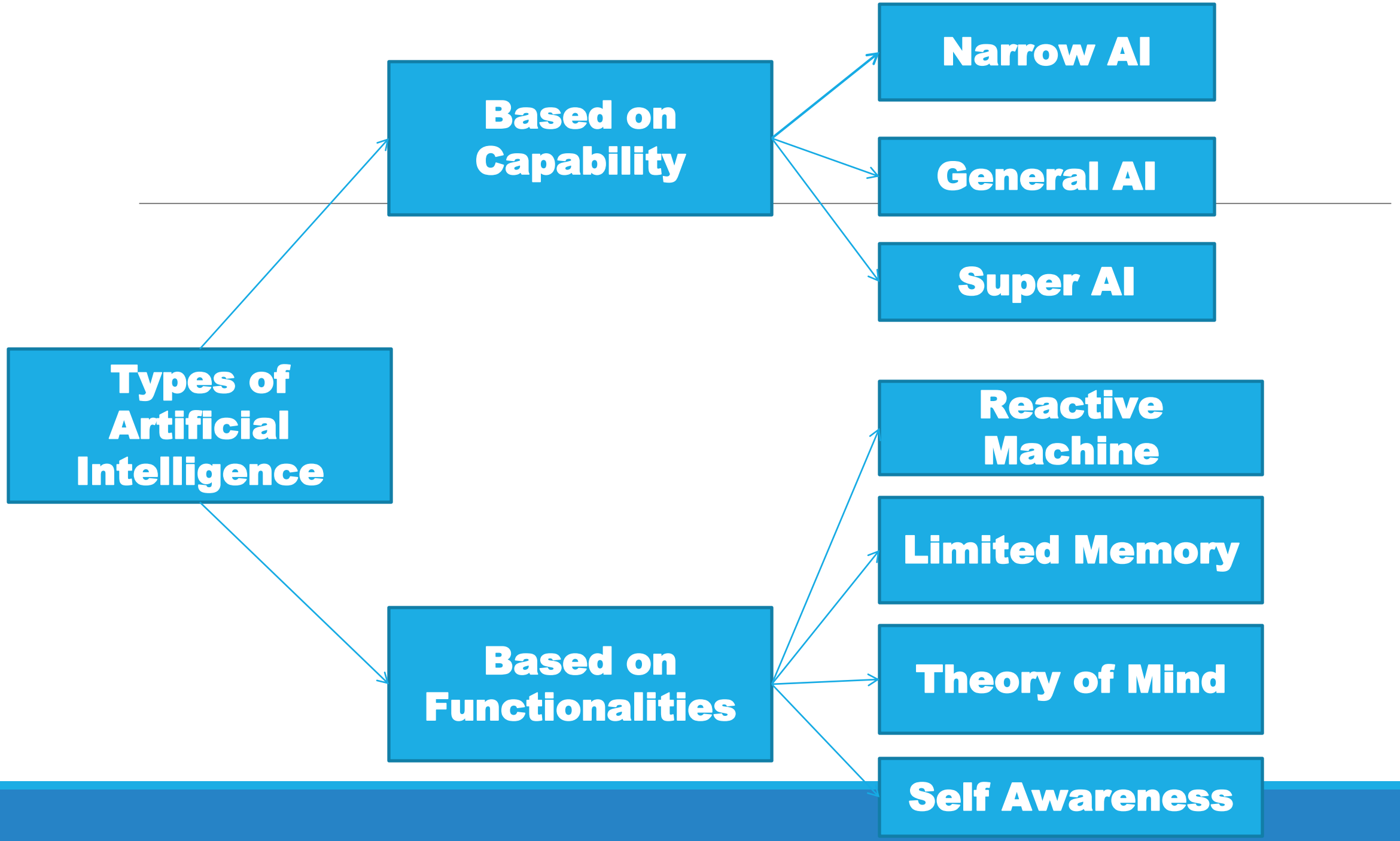
| Game Example | AI Concept Taught |
| --- | --- |
| **AlphaGo** | Deep Learning, Reinforcement Learning, Self-play |
| **Robo Soccer** | Multi-Agent Systems, Path Planning, Real-Time Decision Making |
| **KUKA Robot** | Robotics, Control Systems, Human-Robot Interaction |

# Self Reflection Question

What skills do humans still have that robots don’t?

# Discussion

Creativity, emotion, intuition


Types of Artificial Intelligence
Based on Capability
Narrow AI
General AI
Super AI
Based on Functionalities
Reactive Machine
Limited Memory
Theory of Mind
Self Awareness

# I. Based on Capabilities

| Type | Description | Example |
|---|---|---|
| **1. Narrow AI (Weak AI)** | Can do only specific tasks. Cannot think beyond what it's trained for. | Siri, Alexa, ChatGPT, Google Translate |
| **2. General AI (Strong AI)** | Can understand, learn, and apply knowledge like a human. | Still under research — future goal |
| **3. Super AI** | More intelligent than humans. Can make decisions better and faster. | Fictional – Seen in movies like “Avengers: Age of Ultron” or “Terminator” |

# II. Based on Functionality

| Type | Description | Example |
|---|---|---|
| **1. Reactive Machines** | No memory. Responds only to current input. | IBM’s Deep Blue (Chess) |
| **2. Limited Memory** | Remembers past data temporarily to make decisions. | Self-driving cars, Chatbots |
| **3. Theory of Mind** | Understands emotions, beliefs, intentions. | Still theoretical – future robots |
| **4. Self-aware AI** | Has consciousness, self-awareness, emotions. | Science fiction only – not real yet |

# AI Classification Activity: Classify the AI!

I will show or describe a tool.
You will identify its type:

Capability-wise:

- Narrow AI
- General AI
- Super AI

Functionality-wise:

- Reactive Machines
- Limited Memory
- Theory of Mind
- Self-Aware AI

# Examples to Show or Describe:

| Tool / System | Description | Capability Type | Functionality Type |
|---|---|---|---|
| **Google Maps** | Suggests best routes using traffic and past data | Narrow AI | Limited Memory |
| **JARVIS (Iron Man)** | Talks, learns, plans, controls environment, shows emotions | Super AI (Fictional) | Self-Aware / Theory of Mind |
| **Self-driving Car** | Uses past trips, sensors, and predictions to drive safely | Narrow AI | Limited Memory |
| **Calculator** | Solves arithmetic, no learning or memory | Narrow AI | Reactive Machine |

**Based on strength, breadth, and application**

# AI can be described in different ways

## Weak or Narrow AI

is AI that is applied to specific domain. For example, language translation, virtual assistants, self driving cars, AI-powered web searches, recommendation engines and intelligent spam filters, Applied AI can perform specific tasks, but not learn new ones, making decisions based on programmed algorithms, and training data.

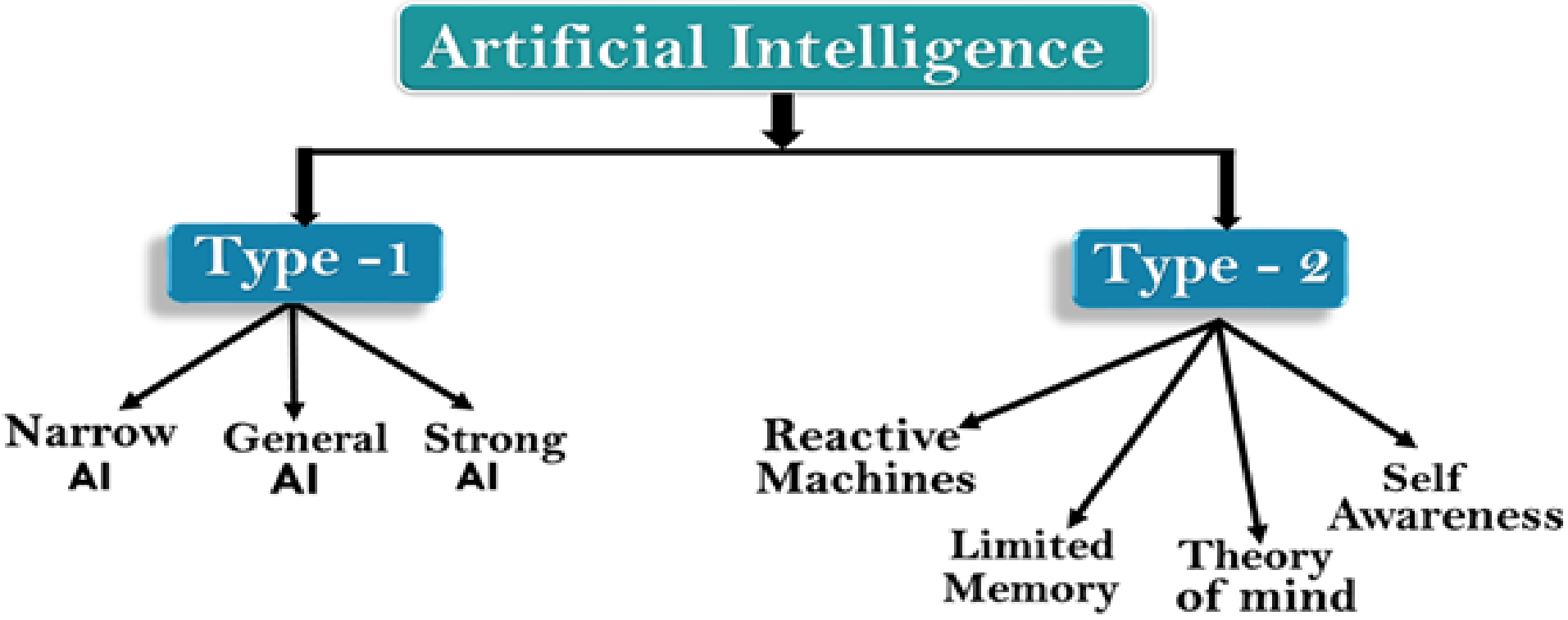
## Strong AI or Generalized AI

is AI that can interact and operate a wide variety of independent and unrelated tasks. It can learn new tasks to solve new problems, and it does this by teaching itself new strategies. Generalized Intelligence is the combination of many AI strategies that learn from experience and can perform at a human level of intelligence

## Super AI or Conscious AI

is AI with human-level consciousness , which would require it to be self-aware. Because we are not yet able to adequately define what consciousness is, it is unlikely that we will be able to create a conscious AI in the near future.

# Types of Artificial Intelligence



# AI type-1: Based on Capabilities

1. Weak AI or Narrow AI:
   - Narrow AI is a type of AI which is able to perform a dedicated task with intelligence.
   - Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task.

# AI type-1: Based on Capabilities

**1. Weak AI or Narrow AI:**

- Apple Siriis a good example of Narrow AI, but it operates with a limited pre-defined range of functions.
- IBM's Watson supercomputer also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.

# AI type-1: Based on Capabilities

1. Weak AI or Narrow AI:
   - Examples of Narrow AI are playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.

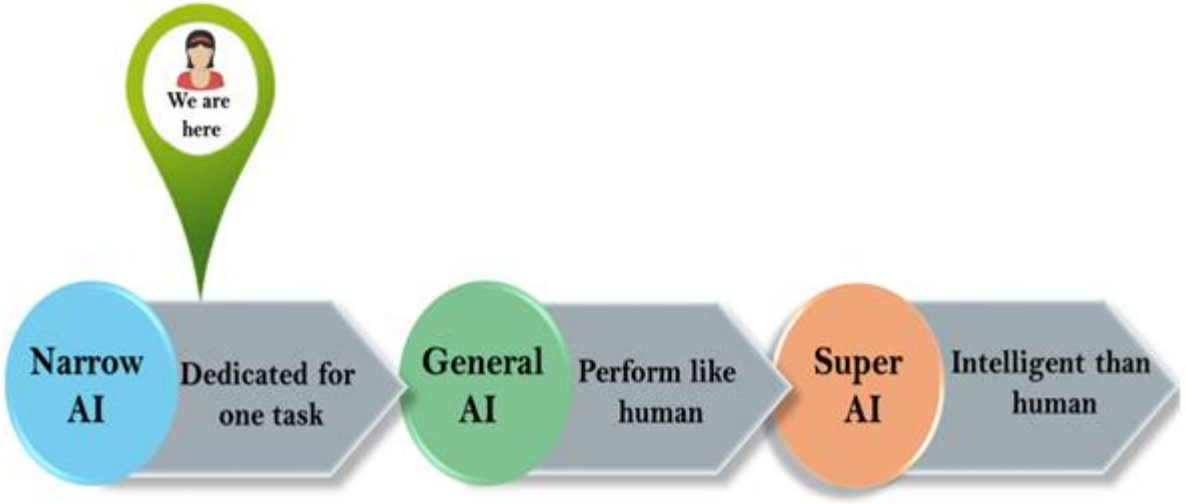
# 2. General AI:

- General AI is a type of intelligence which could perform any intellectual task with efficiency like a human.
- The idea behind the general AI to make such a system which could be smarter and think like a human by its own.
- Currently, there is no such system exist which could come under general AI and can perform any task as perfect as a human.
- As systems with general AI are still under research, and it will take lots of efforts and time to develop such systems.

# 3. Super AI:

- Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties.
- It is an outcome of general AI.
- Some key characteristics of strong AI include capability include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.
- Super AI is still a hypothetical concept of Artificial Intelligence.
- Development of such systems in real is still world changing task.

# AI type-1: Based on Capabilities



# Artificial Intelligence type-2: Based on functionality

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## 1. Reactive Machines

- Purely reactive machines are the most basic types of Artificial Intelligence.
- Such AI systems do not store memories or past experiences for future actions.
- These machines only focus on current scenarios and react on it as per possible best action.
- IBM's Deep Blue system is an example of reactive machines.
- Google's AlphaGo is also an example of reactive machines.

# 2. Limited Memory

- Limited memory machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- Self-driving cars are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

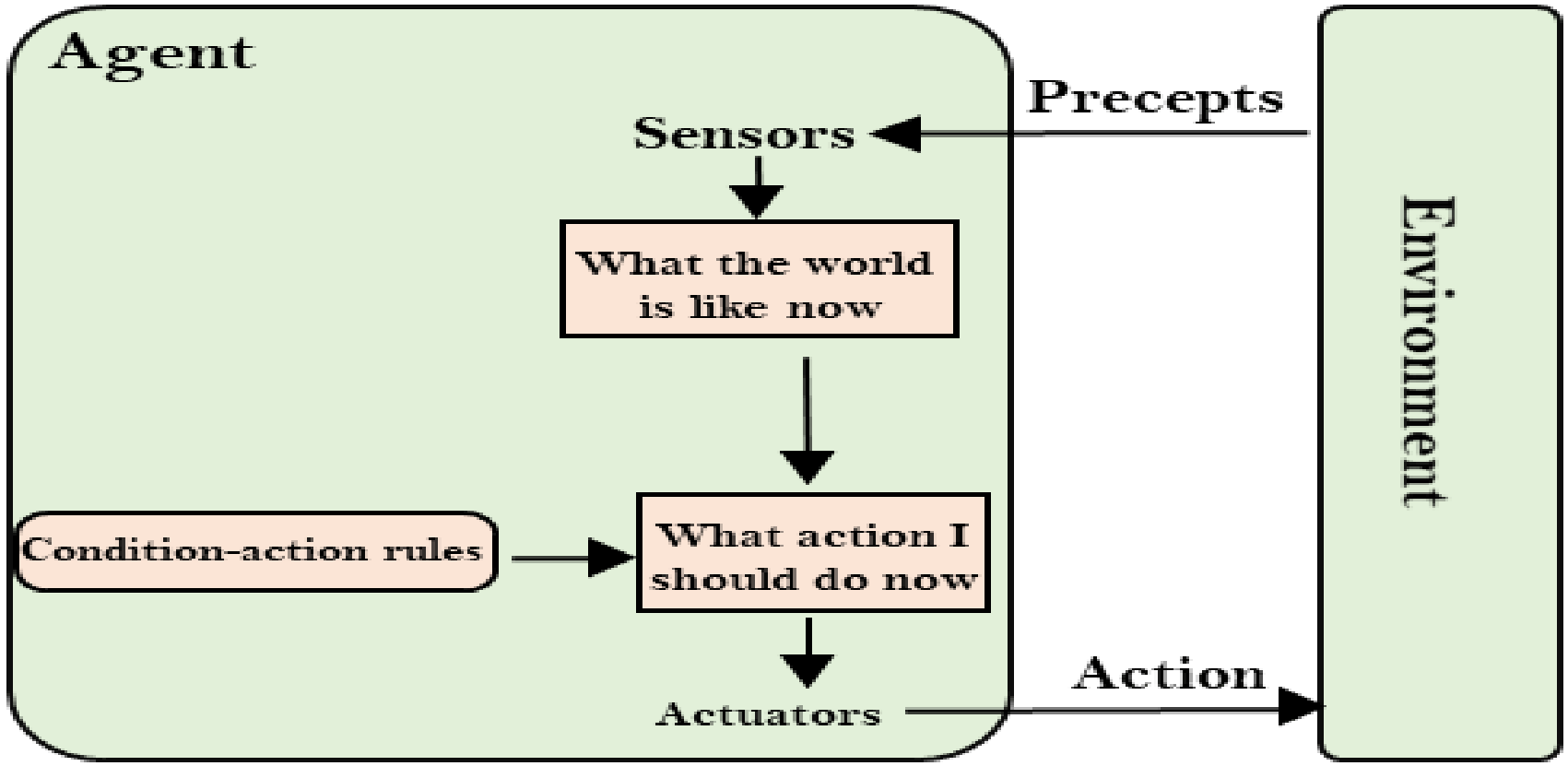
# 3. Theory of Mind

- Theory of Mind AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.
- This type of AI machines are still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.

# 4. Self-Awareness

- Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.
- These machines will be smarter than human mind.
- Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

# AI Agents


Agent
Precepts
Sensors
What the world is like now
Environment
Condition-action rules
What action I should do now
Action
Actuators

# Agents in Artificial Intelligence

An AI system can be defined as the study of the rational agent and its environment. The agents sense the environment through sensors and act on their environment through actuators. An AI agent can have mental properties such as knowledge, belief, intention, etc.

# What is an Agent?

An agent can be anything that perceiveits environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of **perceiving**, **thinking**, and **acting**. An agent can be:

**Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.

# What is an Agent?

**Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.

**Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

# Impact of AI

## AI will Add $15 Trillion To The World Economy By 2030

- AI is no longer science fiction
- Impacting multiple industries and human beings on daily basis
- AI contributed a whopping $2 trillion to global GDP last year
- By 2030, it could be as much as $15.7 trillion, “making it the biggest commercial opportunity in today’s fast changing economy,”

https://www.forbes.com

# Turing Test:

During the Turing Test, the human interrogator asks several questions to both players. Based on the answers, the interrogator attempts to determine which player is a computer and which player is a human respondent.

Player A
**Computer**





Player B
**Human Responder**



Player C
**Interrogator**

■ Question to Respondents

■ Answers to Question

abc NEWS



Thinking Humanly: The cognitive modeling approach

Thinking Rationally: The laws of thought approach

Artificial Intelligence

Acting Humanly: The Turing Test approach

Acting Rationally: The rational agent approach